Maximum Likelihood Learning & Latent Variable Models

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Generative Modeling DSAIT4030, 29/04/2025

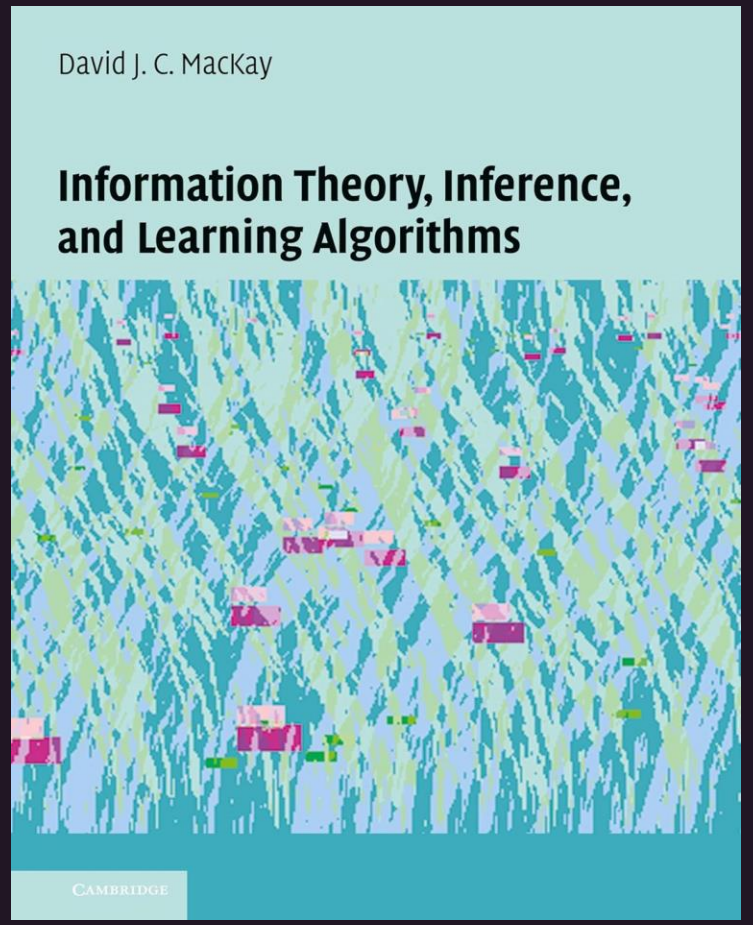
Making your dream indistinguishable from reality

Finding a learning objective

Recall:

"a *generative model* describes a process that is assumed to give rise to some data"

- David J.C. MacKay



Finding a learning objective

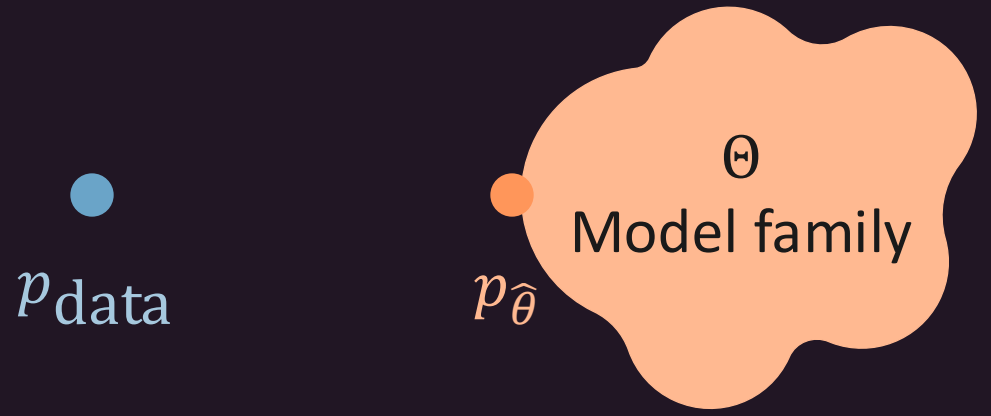
We want to find a good $\hat{\theta} \in \Theta$ such that $p_{\hat{\theta}} \approx p_{\text{data}}$ (we assume we have a dataset $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$ with N samples from p_{data})

Finding a learning objective

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$\mathbf{x}_i \sim p_{\text{data}}$
$i = 1,2, \ldots, N$

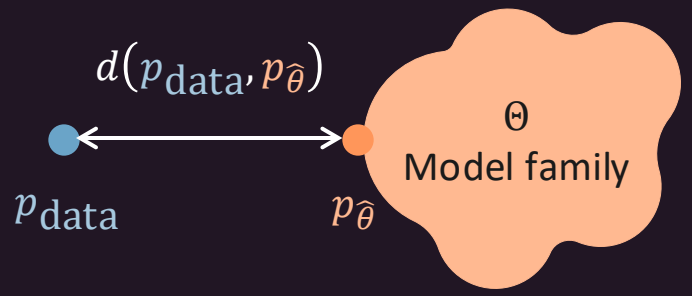


Cat images from RedBubble

Finding a learning objective



$\mathbf{x}_i \sim p_{\text{data}}$
$i = 1,2,\ldots,N$



💡 define an objective $d(p_{\text{data}}, p_\theta)$ that captures 'closeness' between p_θ and p_{data}, and choose the model closest to the truth

$$\hat{\theta} \in \arg\min_{\theta} d(p_{\text{data}}, p_\theta)$$

Cat images from RedBubble

The weight of evidence

$$\frac{\Pr(P|D)}{\Pr(Q|D)}$$

The weight of evidence

$$\frac{\Pr(P|D)}{\Pr(Q|D)} = \frac{\Pr(D|P)}{\Pr(D|Q)} \frac{\Pr(P)}{\Pr(Q)}$$

The weight of evidence

$$\log \frac{\Pr(P|D)}{\Pr(Q|D)} = \log \frac{\Pr(D|P)}{\Pr(D|Q)} + \log \frac{\Pr(P)}{\Pr(Q)}$$

The weight of evidence

$$\log \frac{\Pr(P|D)}{\Pr(Q|D)} = \log \frac{p(D)}{q(D)} + \log \frac{\Pr(P)}{\Pr(Q)}$$

Now imagine if one of the hypotheses were true

$$\mathrm{KL}[p; q] := \int p(x) \log \frac{p(x)}{q(x)} dx = \mathbb{E}_p \left[\log \frac{p(x)}{q(x)} \right]$$

If $p$was the distribution governing the world, KL Divergence is the *expected weight of evidence* we would accumulate from observing data

How quickly would we be able to discern between hypotheses P and Q if Pwere true.

Adapted from "KL is all you need" – Alex Alemi (blog.alexalemi.com KL is All You Need)

Properties of KL Divergence

- KL is asymmetric

$$\mathbb{E}_p\left[\log \frac{p(x)}{q(x)}\right] \neq \mathbb{E}_q\left[\log \frac{q(x)}{p(x)}\right]$$

- KL is zero iff $p = q$ (if and only if)

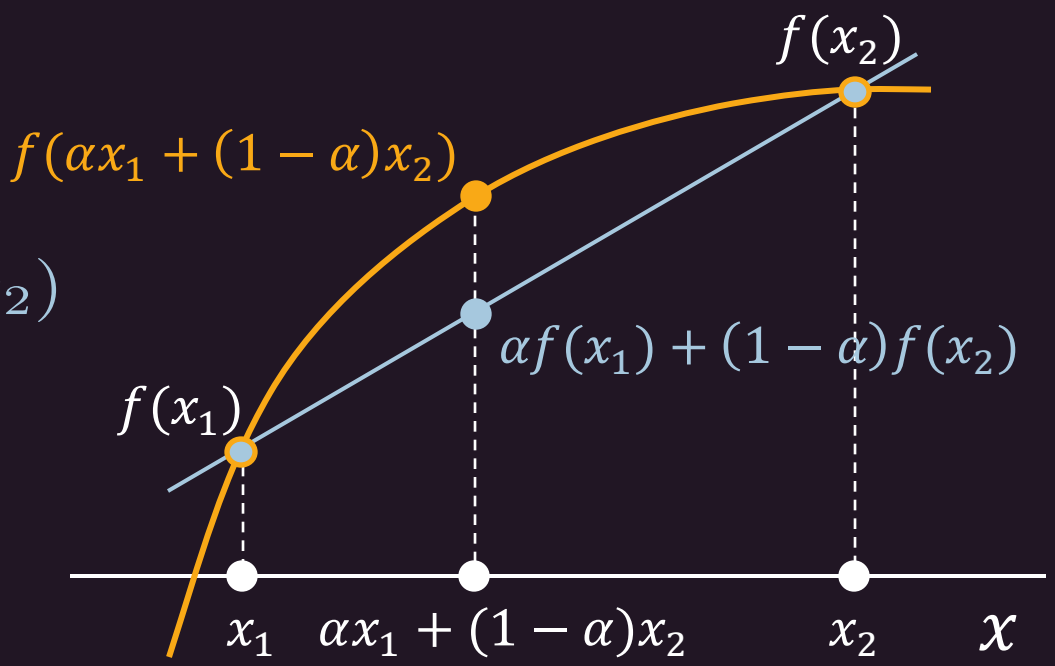
$$\mathbb{E}_p\left[\log \frac{p(x)}{q(x)}\right] = \mathbb{E}_p\left[\log \frac{p(x)}{p(x)}\right] = 0$$

Recap: Jensen's Inequality and concave functions

For a concave function f, we have $f(\sum_{i=1}^{n} \alpha_i \boldsymbol{x}_i) \geq \sum_{i=1}^{n} \alpha_i f(\boldsymbol{x}_i)$, where $\alpha_i \geq 0$ and $\sum_{i=1}^{n} \alpha_i = 1$.

$$f(\alpha \boldsymbol{x}_1 + (1-\alpha)\boldsymbol{x}_2) \geq \alpha f(\boldsymbol{x}_1) + (1-\alpha) f(\boldsymbol{x}_2)$$

$$f(\mathbb{E}[\boldsymbol{x}]) \geq \mathbb{E}[f(\boldsymbol{x})]$$



Properties of KL Divergence

- KL is non-negative

$$\mathbb{E}_p \left[\log \frac{p(x)}{q(x)} \right] \geq 0$$

Proof:

$$\begin{aligned}
\mathrm{KL}(p||q) &:= \int p(x) \left(-\log \frac{q(x)}{p(x)} \right) dx \\
&\geq -\log \int p(x) \frac{q(x)}{p(x)} \quad \text{(Jensen's Inequality)} \\
&= -\log \int q(x) dx \\
&= 0
\end{aligned}$$

Properties of KL Divergence

- KL is non-negative

$$\mathbb{E}_p \left[\log \frac{p(x)}{q(x)} \right] \geq 0$$

- KL is monotonic, follows from a generalized version of data processing inequality (*post-processing cannot increase information*)

$$\mathbb{E}_p \left[\log \frac{p(x,z)}{q(x,z)} \right] = \mathbb{E}_p \left[\log \frac{p(z|x)}{q(z|x)} + \log \frac{p(x)}{q(x)} \right] \geq \mathbb{E}_p \left[\log \frac{p(x)}{q(x)} \right] \geq 0$$

From KL divergence to log-likelihood

Let us revisit the learning objective using KL divergence as a distance

$$\hat{\theta} \in \arg\min_{\theta} \mathrm{KL}(p_{\mathrm{data}}, p_\theta)$$

Simplifying,

$$\begin{aligned}\mathrm{KL}(p_{\mathrm{data}}, p_\theta) &= \mathbb{E}_{\mathbf{X}\sim p_{\mathrm{data}}}\left[\log \frac{p_{\mathrm{data}}(\mathbf{X})}{p_\theta(\mathbf{X})}\right] \\ &= \underbrace{\mathbb{E}_{\mathbf{X}\sim p_{\mathrm{data}}}[\log p_{\mathrm{data}}(\mathbf{X})]}_{\text{does not depend on } \theta} - \underbrace{\mathbb{E}_{\mathbf{X}\sim p_{\mathrm{data}}}[\log p_\theta(\mathbf{X})]}_{\text{expected log-likelihood}}\end{aligned}$$

Minimizing KL divergence is equivalent to maximizing likelihood!

From KL divergence to log-likelihood

minimizing KL is equivalent to maximizing likelihood

$$\arg\min_\theta \mathrm{KL}(p_{\mathrm{data}}, p_\theta) = \arg\min_\theta -\mathbb{E}_{\mathbf{x}\sim p_{\mathrm{data}}}[\log p_\theta(\mathbf{X})]$$
$$= \arg\max_\theta \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{data}}}[\log p_\theta(\mathbf{X})]$$

- 😖 We do not know p_{data}, so we can't compute the expectation
- 😎 We have samples from p_{data}
- 💡 Use the law of large numbers to approximate the expectation with the empirical distribution:

$$\mathbb{E}_{p_{\mathrm{data}}}[\log p_\theta(\mathbf{X})] \approx \underbrace{\frac{1}{n}\sum_{i=1}^{n} \log p_\theta(\mathbf{x}_i) = l(\theta)}_{\text{log-likelihood function}}$$

A universal recipe for objectives

Universal recipe

1. Draw the causal model of the real world (P)
2. Augment the real world with any components you wish to add
3. Rewire the diagram to reflect your desires, your dream world (Q)
4. Minimize $KL[P;Q]$
5. …
6. Profit!

"KL is all you need" – Alex Alemi (blog.alexalemi.com KL is All You Need)

Density estimation using the universal recipe

World P
world we have

$p(x)$



Images

World Q
world we want

$q_\theta(x)$



Images

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Density estimation using the universal recipe

$$\underbrace{\mathbb{E}_p\left[\log \frac{p(\mathbf{X})}{q_\theta(\mathbf{X})}\right]}_{KL[p;q]} = \underbrace{\mathbb{E}_p[\log p(\mathbf{X})]}_{-H[p]} + \underbrace{\mathbb{E}_p[-\log q_\theta(\mathbf{X})]}_{H[p;q]}$$

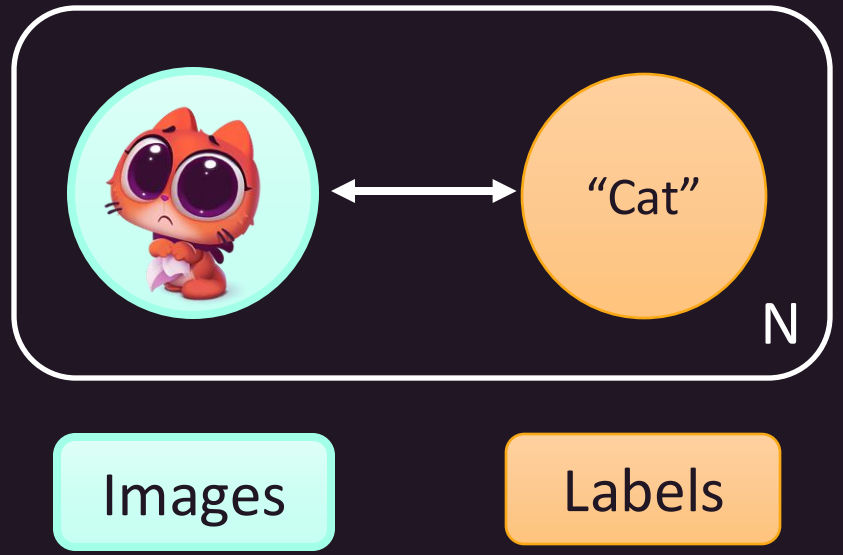
Cross-entropy is KL minimization

"KL is all you need" – Alex Alemi (blog.alexalemi.com KL is All You Need)

Supervised Learning

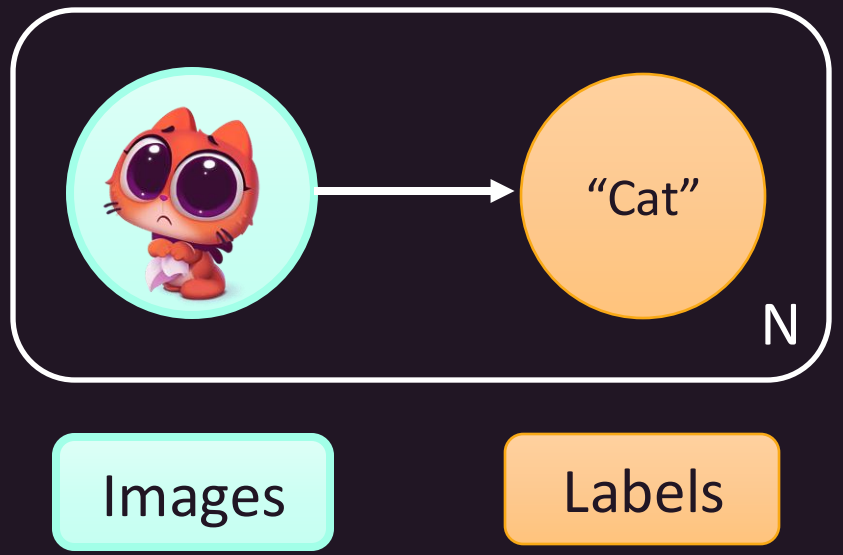
World P
world we have

$$p(x, y)$$



World Q
world we want

$$q(x, y) = p(x)q_\theta(y|x)$$



"KL is all you need" – Alex Alemi (blog.alexalemi.com KL is All You Need)

Supervised Learning

$$\mathbb{E}\left[\log \frac{p(\mathbf{X})p(\mathbf{y}|\mathbf{X})}{p(\mathbf{X})q_\theta(\mathbf{y}|\mathbf{X})}\right] = \mathbb{E}\left[\log \frac{p(\mathbf{y}|\mathbf{X})}{q_\theta(\mathbf{y}|\mathbf{X})}\right] \geq 0$$

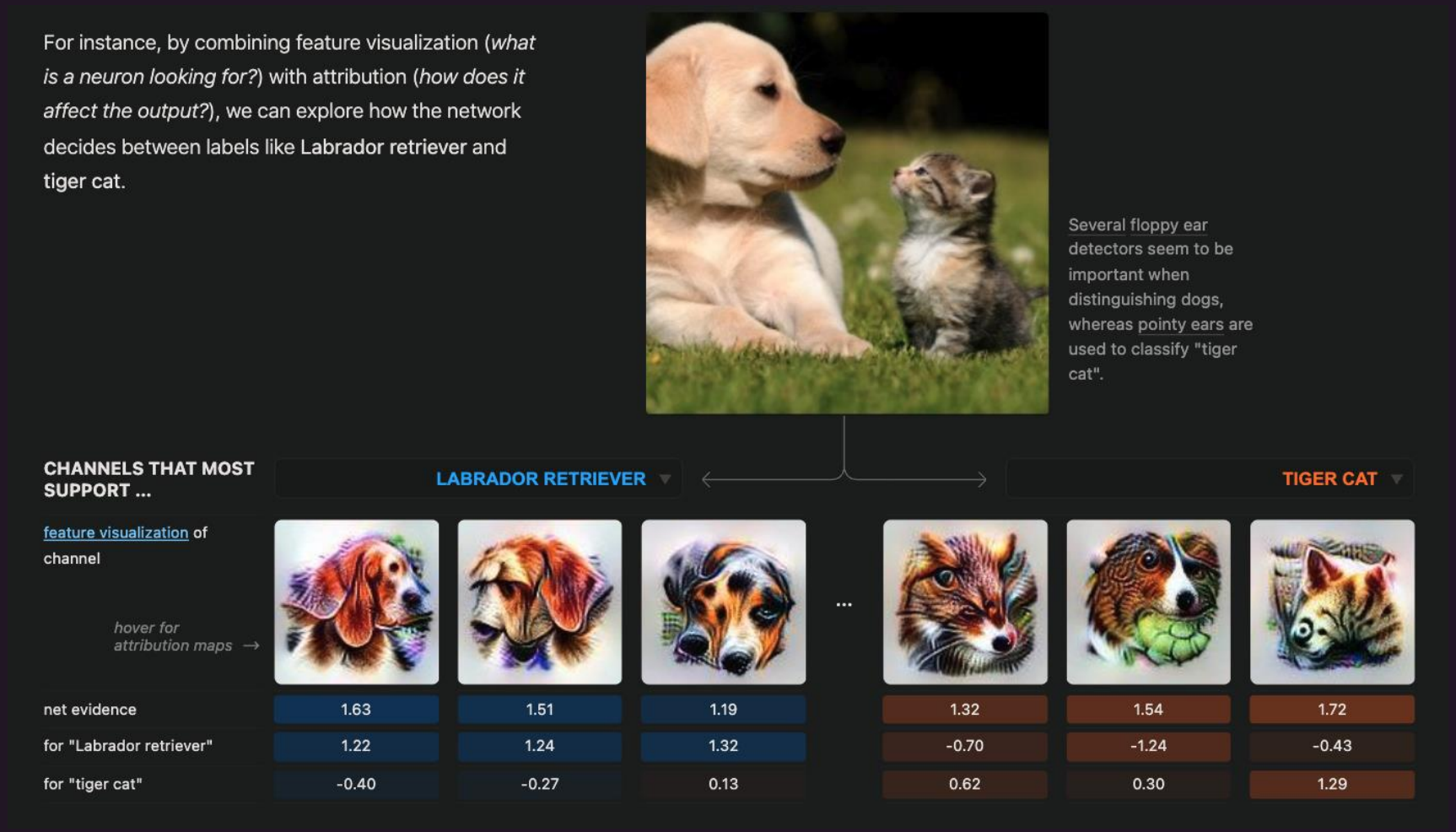
$$\mathbb{E}_p\left[-\log q_\theta(\mathbf{y}|\mathbf{X})\right] \geq \mathbb{E}_p\left[-\log p(\mathbf{y}|\mathbf{X})\right]$$

Cross entropy

Data entropy

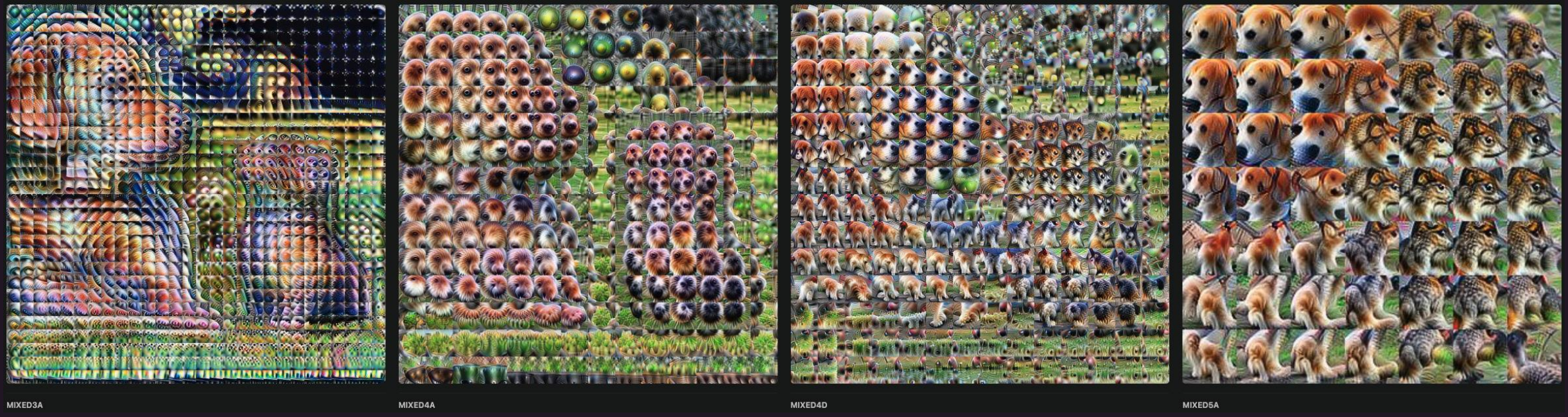
"KL is all you need" – Alex Alemi (blog.alexalemi.com KL is All You Need)

Intermediate layers – a happy accident



“"The Building Blocks of Interpretability"” – Olah, et al. , Distill 2018. https://doi.org/10.23915/distill.00010

Intermediate layers – a happy accident



How do we learn useful representations?

“"The Building Blocks of Interpretability"” – Olah, et al. , Distill 2018. https://doi.org/10.23915/distill.00010

Latent Variable Models

Motivation

Cat memes $\mathbf{X}$ can feature a lot of interesting characteristics: fur color, breed, pose, expression, etc.



⚠ These factors of variation (latent), unless annotated, are not available

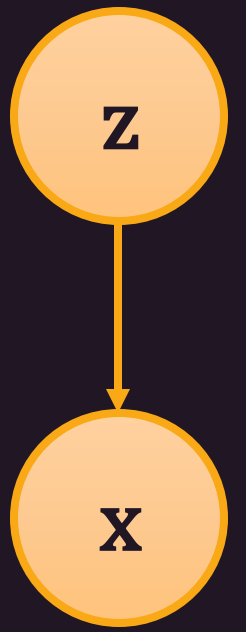
❓ How to automatically learn these characteristics from data?

💡 Explicitly model these factors as latent variables Z and learn them using unsupervised learning

Definition

A latent variable model defines a probability distribution

$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{x}|\mathbf{z})p(\mathbf{z})$$



The joint is also described by the probabilistic graphical model; a connection denotes stochastic dependence

Observed variables $\mathbf{x}$ represent high-dimensional objects that we are directly available

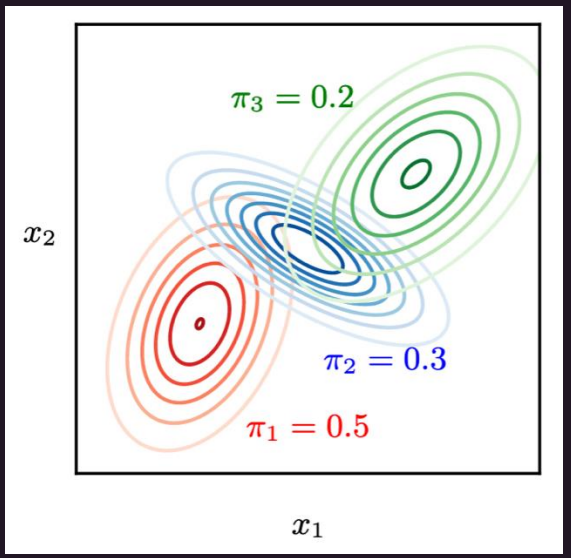
Latent variables $\mathbf{z}$ are the unobserved factors associated with $\mathbf{x}$ as specified by the joint distribution $p(\mathrm{x}, \mathrm{z})$

Discrete Latents: Gaussian Mixture Models (GMMs)

Assumption: data is a convex combination of K Gaussian distributions

Consider a K–dimensional binary random variable $\mathbf{z}$ with a 1-of-K representation; we introduce mixing coefficients π_k, such that

$$p(z_k = 1) = \pi_k; \quad 0 \leq \pi_k \leq 1; \quad \sum_{k=1}^{K} \pi_k = 1$$



The marginal distribution is $$p(\mathbf{z}) = \prod_{k=1}^{K} \pi_k^{z_k}$$

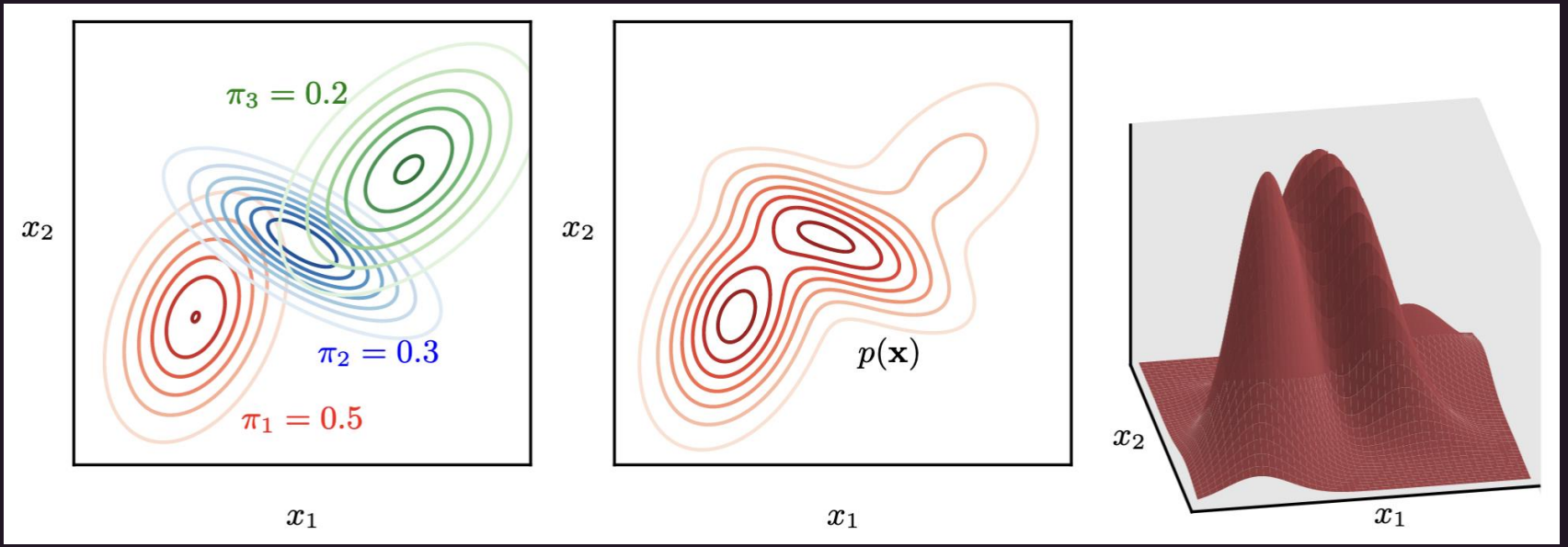
The conditional is a Gaussian, $$p(\mathbf{x}|z_k = 1) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

"Deep Learning: Foundations and Concepts" – Chris Bishop (https://www.bishopbook.com/)

Discrete Latents: Gaussian Mixture Models (GMMs)

Taken together, the marginal distribution of $\mathbf{x}$ is

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z})p(\mathbf{x}|\mathbf{z}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$



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Gaussian Mixture Models: Learning

For a dateset $\mathbf{X}$ of *iid* observations $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$, fit a GMM by optimizing the log-likelihood function

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

wrt the model parameters $\{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^{K}$.

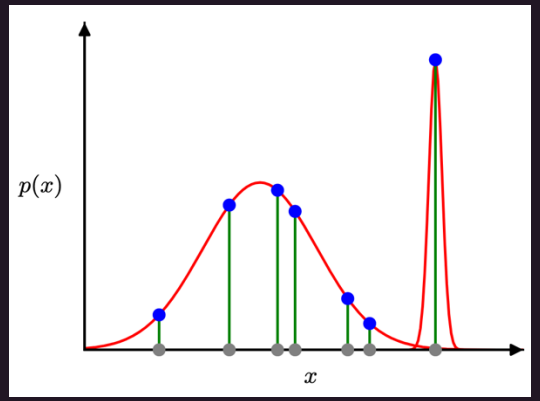
In principle, for a large enough K and N, a mixture of Gaussians can model (approximate) any distribution

Gaussian Mixture Models: Learning

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

Challenges?

- 😕 Summation inside the logarithm -> no closed-form solution
- 😣 Singularities! Assume $\boldsymbol{\Sigma}_k = \sigma_k^2 \mathbf{I}$ If $\sigma_j \to 0, \boldsymbol{\mu}_j = \mathbf{x}_n$, log-likelihood goes to infinity



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- 😐 Non-identifiability. $K!$ equivalent solutions for a K–component mixture

Gaussian Mixture Models: Learning

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

We can use gradient-based techniques (e.g. SGD) with the log-sum-exp trick for numerical stability:

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \exp(\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)) \right\}$$

Challenge:
Maintain constraints on π_k and $\boldsymbol{\Sigma}_k$ (covariance matrices are positive semidefinite)

So what can we do?

If z_n is known for each x_n, then the summation inside the ln disappears.

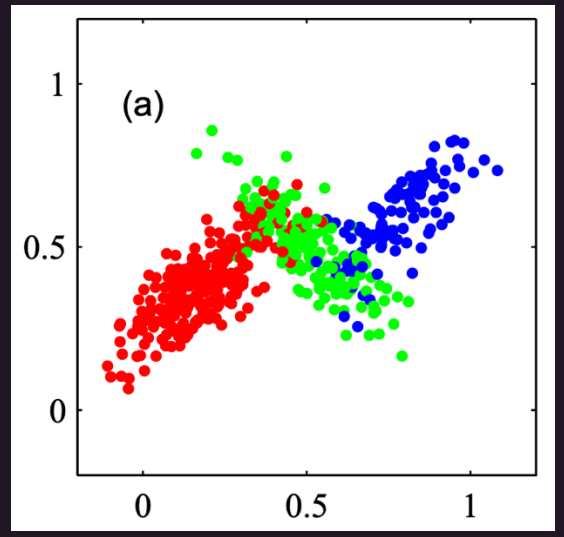
- When z_n is known, the complete log-likelihood is

$$\ell(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln p(\mathbf{x}_n, \mathbf{z}_n | \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln p(\mathbf{x}_n | \mathbf{z}_n; \boldsymbol{\mu}, \boldsymbol{\Sigma}) p(\mathbf{z}_n | \boldsymbol{\pi})$$

- When z_n is unknown, the log-likelihood is

$$\ell(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \sum_{k=1}^{K} p(\mathbf{x}_n, z_k = 1 | \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$= \sum_{n=1}^{N} \ln \sum_{k=1}^{K} p(\mathbf{x}_n | z_k = 1; \boldsymbol{\mu}, \boldsymbol{\Sigma}) p(z_k = 1 | \boldsymbol{\pi})$$



"Deep Learning: Foundations and Concepts" – Chris Bishop (https://www.bishopbook.com/)

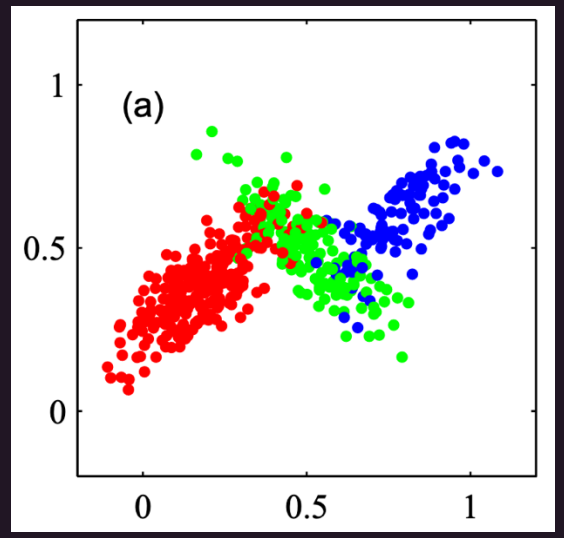
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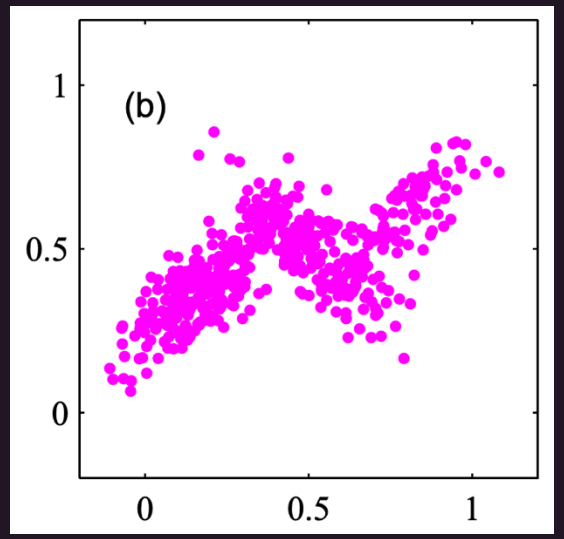
This is a convex optimization problem for GMMs,
see Bishop Sec. 15.2.2 for derivation

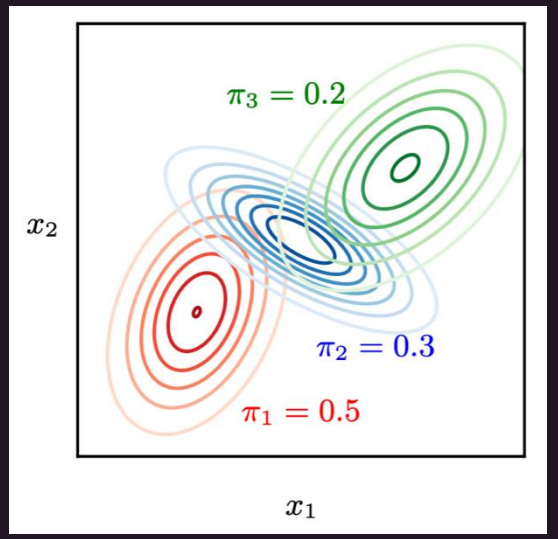


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So what can we do?

When z_n is unknown, but parameters are known, computation is easy





$$p(z_k = 1|\mathbf{x}) = \frac{p(z_k = 1)p(\mathbf{x}|z_k = 1)}{\sum_{j=1}^{K} p(z_j = 1)p(\mathbf{x}|z_j = 1)} = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

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We perform iterative optimization!

The Expectation Maximization (EM) algorithm is a powerful technique for finding maximum likelihood solutions for latent variable models

E Step

- Solve the inference problem: *which Gaussian generated each datapoint?* or more precisely, *how responsible is each Gaussian for generating each datapoint?*
- Compute the posterior probabilities, $p(z_k = 1|\mathbf{x})$

M Step

- Use the assigned *responsibilities* (the posterior probabilities from the E step) to re-estimate the means, covariances, and mixing coefficients

ELBO: Toward a general perspective on the EM

Consider a probabilistic model with all observed variables $\mathbf{X}$ and latent variables $\mathbf{Z}$. Introducing a distribution $q(\mathbf{Z})$, we can write:

$$\ln p(\mathbf{X}|\boldsymbol{\theta}) = \ln \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})$$

(assuming discrete $\mathbf{Z}$, derivation is identical for continuous $\mathbf{Z}$ with integration replacing summation)

$$= \ln \sum_{\mathbf{Z}} \frac{q(\mathbf{Z})}{q(\mathbf{Z})} p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})$$

$$= \ln \mathbb{E}_q \left[\frac{p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})}{q(\mathbf{Z})} \right]$$

$$\geq \mathbb{E}_q \left[\ln \frac{p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})}{q(\mathbf{Z})} \right]$$

… Jensen's Inequality

$$= \mathcal{L}(q, \boldsymbol{\theta})$$

… Evidence Lower Bound, or ELBO

If you cannot answer a hard question directly,
reframe it into an easier question
you can (hopefully) answer exactly

—